

SPECTRAVIDEO SV-318



The Spectravideo advertisements trumpet "Some computers are better than others. Ours is *much* better". They could be right about that, although I am not convinced about some of the comparisons they make, as they include or exclude extra-cost items to suit themselves.

So, what does the SV-318 give you for your money? There are a lot of nice things, plus a nasty rubber keyboard. The main problem is that you often get tactile feedback even though the keystroke hasn't registered — it is essential to press the centre of the key. A 'keyclick' can be generated electronically and this makes typing much easier, but tends to annoy other people in the room. There are five shiftable function keys which are genuinely user-programmable and therefore very useful when exploring particular features of the machine or when programming. A built-in joystick sits to the left of the keyboard proper, and the stick itself is removable for those who prefer a joypad. I was amused by the provision of two sticks, presumably in case of loss. One or two external joysticks may be added to the Atari-style connectors on the right-hand side of the case. Games enthusiasts are also served by the cartridge socket hidden beneath a spring-loaded flap above the keyboard. Unfortunately no cartridges were received.

The display was very good, with clear characters and graphics as well as nice solid colors. The text screen is a nominal 40 columns, but the default is 39, presumably to cater for TV screens with excessive overscan. The 24th line of the display may be used to label the function keys. There are two graphics resolutions: 256 x 192 and 64 x 48, both giving 15 colors plus a special 'transparent' code giving the current background color. The SV-318 claims to have 'sprite' graphics, but my understanding is that one of the attributes of sprites is that they have velocity, that is to say once set in motion they continue to move without explicit programming. The Spectravideo sprites lack this ability: they are essentially programmable shapes that can pass in front of or behind each other. They still simplify the programming of animated displays, and the easy detection of collisions helps arcade game designers.

The 80 column capability mentioned in the advertisement is obtained by adding an 80 column card, but the expansion is a prerequisite. I believe purchasers will appreciate the switch box included with the system which selects antenna or computer input to the television without unplugging leads. The leads supplied with the SV-318 are reasonably long.

The version of Microsoft Basic that is built into the Spectravideo is very similar to that used on the Tandy Color Computer and the Hitachi Peach, as well as the GW-Basic/Basic-A found on increasing numbers of 16-bit systems. The extensions cover graphics and other hardware features like sound and joysticks. The 'macro languages' for music and graphics are part of the implementation, but sound effects are not so easy to program: the SOUND command allows the user to load any byte into any register of the sound generator, but this is hardly better than the POKE or OUT commands that would otherwise be used. A screen editor is included, and benefits greatly from having the cursor position controlled by the joystick — it makes editing quick and easy.

The very powerful interrupt feature has been included, with interrupts being generated by the stop and function keys, joystick triggers (the space bar serves this purpose for the built in joystick), sprite collisions, and a 1/60th second timer. There are also commands for switching memory banks, but the manuals do not make clear what use can be made of this feature. Further commands allow the use of the audio channel of the Spectravideo cassette deck. This could be useful in educational programs, but again there was no cassette deck.

Good documentation is particularly important to buyers of home computers, as (unlike a business user) they cannot turn to consultants or support staff. The tutorial manual supplied with the SV-318 is fine as far as it goes, but it does little more than introduce the concepts of Basic programming and the features of the Spectravideo. I am particularly concerned about the way sprites are treated: the concepts involved are not brought out in the discussion and readers are left to discover the principles for themselves. 'Learning by doing' may have its advantages, but I would expect the information to be included somewhere in the documentation. Some hints may be found in the quick reference guide, but surely this should only be a memory-jogger? The problem comes down to the lack of a proper reference manual, so perhaps someone will publish "How to get the most from your Spectravideo" in time for Christmas (to any author working on such a book: *please* include a good index).

Overall, the SV-318 provides plenty of features at a modest price. The importers say that most of the optional extras are already available, although as the more important ones require the expansion box this should be budgeted for. The version of Basic used is powerful, yet very similar to those used on other machines. If it weren't for the keyboard, it would probably be on my wish-list.

Processor	Z80
RAM	32k (approx 12½k usable)
ROM	32k
Ease of use	★ ★ ★
Ease of programming	★ ★ ★ ★ ★
Expansion and accessories	★ ★ ★ ★
Documentation	★ ★
Presentation	★ ★ ★ ★

*Review machine supplied by VideoActiv Electronics,
St. Kilda, Vic.*